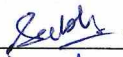
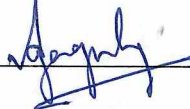


PART-B

Name of the Student: Subhashree Kedia

Signature of the Student: 

Roll No.: 230123063

Signature of the Invigilator: 

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	3	2	5	1	11

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX} \\ S_{yX}^T & S_{XX} \end{pmatrix} \quad S_{yX} \text{ is } 1 \times q$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

we apply Canonical representation theorem here,

the eigenvalues of $M = S_{yy}^{-1} S_{yx} S_{xx}^{-1} S_{xy}$ give the squared canonical correlations.

Here S_{yy} is a single value because $Y_{1 \times 1}$

$$M_{1 \times 1} = \frac{S_{yx} S_{xx}^{-1} S_{xy}}{S_{yy}} \text{ is also a } 1 \times 1 \text{ matrix.}$$

$$\text{eigenvalue value of } M = \text{value of } \left[\frac{S_{yx} S_{xx}^{-1} S_{xy}}{S_{yy}} \right]$$

$$\Rightarrow r_1^2 = (\text{first canonical correlation})^2$$

$$\Rightarrow r_1^2 = \frac{S_{yx} S_{xx}^{-1} S_{xy}}{S_{yy}}$$

from multiple linear regression of Y on vector X ,

$$R^2 = \frac{SS_{\text{Reg}}}{SS_T}$$

$$SS_T = \sum (y_i - \bar{y})^2 = S_{yy}$$

$$SS_{\text{Reg}} = \sum (\hat{y}_i - \bar{y})^2$$

$$x_1 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + x_2 \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

For $x_1 = 1, x_2 = 0$ we get $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$

and $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
 - (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
 - (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

~~$\tilde{\mathbf{L}} = \tilde{\mathbf{A}}_2$~~

(a) Ordering in terms of eigenvalues $\rightarrow \lambda_3, \lambda_2, \lambda_1, \lambda_4$

$m=2$, so we will pick λ_3 & λ_2 to ~~make~~ make $\tilde{\mathbf{L}}$.
using the principal component method.

$$\tilde{\mathbf{L}}_{4 \times 2} = [\sqrt{\lambda_3} \mathbf{e}_3 \mid \sqrt{\lambda_2} \mathbf{e}_2]$$

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$$\tilde{\mathbf{L}}_{4 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & \sqrt{2.1} \\ \sqrt{4.2} & 0 \\ 0 & 0 \end{bmatrix} \quad \times$$

(b) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$

Estimated communality of the first variable here
 $\tilde{h}_1^2 = \tilde{l}_{11}^2 + \tilde{l}_{12}^2 = (0.8)^2 + (0.4)^2$

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$$\Rightarrow \tilde{h}_1^2 = 0.8$$

~~Q.2~~

(C). If we use the principal component method,

the $\tilde{L}_{p \times 2} = [\sqrt{\lambda_i} e_i | \sqrt{\lambda_j} e_j]$ where λ_i & λ_j are the largest $m=2$ eigenvalues of S , ($i \neq j$ obviously)

$\Rightarrow \tilde{L} = \begin{bmatrix} 0 & \vdots & \sqrt{\lambda_j} \\ \vdots & \sqrt{\lambda_i} & \vdots \\ \vdots & \vdots & 0 \end{bmatrix}$
 $\leftarrow$ i th row $\leftarrow$ j th row.

As we know the estimated communality,

$$\tilde{h}_i^2 = \sum_{k=1}^m \tilde{l}_{ik}^2 = \tilde{l}_{i1}^2 + \tilde{l}_{i2}^2$$

$$\tilde{h}_k^2 = 0 + 0 = 0 \quad \forall k \neq i \& k \neq j \& 1 \leq k \leq p$$

$$\tilde{h}_i^2 = \lambda_i \quad \& \quad \tilde{h}_j^2 = \lambda_j$$

$$\sum_{k=1}^p \tilde{h}_k^2 = \sum_{\substack{k=1 \\ i \neq k \\ i \neq j}}^{p-m} \tilde{h}_k^2 + \tilde{h}_i^2 + \tilde{h}_j^2 = \lambda_i + \lambda_j$$

$\hat{=}$ Sum of largest 2 eigenvalues of S .

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Hence, proved.

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

~~Let~~ let the transformation be $g(\bar{X}_n)$

mean of $\bar{X}_n = \mu$ & asymptotic variance of $\bar{X}_n = \frac{\mu^2}{n}$
(By central limit theorem).

Using Delta method,

$$g(\bar{X}_n) = g(E\bar{X}_n) + (\bar{X}_n - E\bar{X}_n) g'(E\bar{X}_n)$$

$$\begin{aligned} \text{Var}(g(\bar{X}_n)) &= \text{Var}(\bar{X}_n - E\bar{X}_n) \times [g'(E\bar{X}_n)]^2 \\ &= \text{Var}(\bar{X}_n) \times [g'(E\bar{X}_n)]^2 \end{aligned}$$

$$\begin{aligned} \text{asymptotic variance of } g(\bar{X}_n) &= [g'(E\bar{X}_n)]^2 \times \text{asy. var. of } \bar{X}_n \\ &= [g'(E\bar{X}_n)]^2 \times \frac{\mu^2}{n} \end{aligned}$$

$$\therefore [g'(\bar{X}_n)]^2 \times \frac{\mu^2}{n} = \text{constant for the given condition.}$$

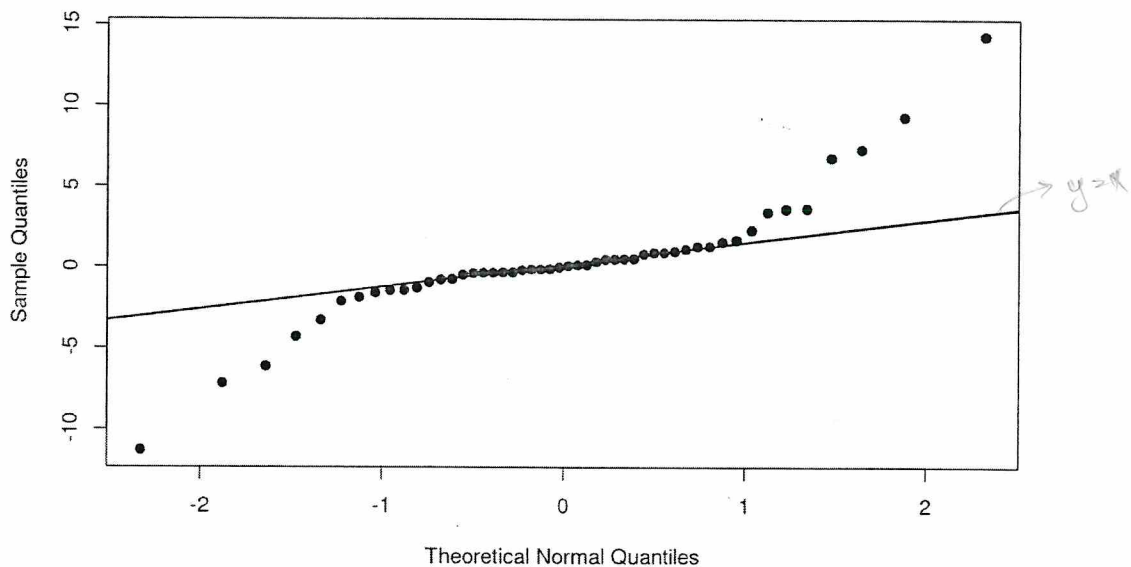
$$\text{Let } g'(\bar{X}_n) = \sqrt{\frac{C_1}{\mu^2}} = \frac{C_1}{\mu}$$

$$\Rightarrow g(\bar{X}_n) = \int \frac{C_1}{\mu} = C_1 \ln(\mu) + C_2$$

So, variance stabilising transformation $g(\bar{X}_n) = \ln(\mu) + C_2$

for some constants
 $C_1, C_2 > 0$

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



The middle part of the distribution follows normal distribution. The right & left extremes show deviation from the normal distribution which means we have more points away from the mean than we should have had in a normal distribution.

So, the distribution is non-normal.

Uniform distribution better categorises this type of data.

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ADDITIONAL PAGES FOR ANSWERS

